

Datalink Security in the L-Band Digital Aeronautical Communications System (LDACS) for Air Traffic Management

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INTRODUCTION

Today's very high-frequency (VHF) voice-based air-ground communication system for tactical aircraft guidance is suffering from the VHF band's increasing saturation in high-density areas [1]. The air-ground¹ communication infrastructure is therefore undergoing modernization to ensure the sustainable growth of the European air transportation system in the coming decades [2], [3]. One major goal of this effort is the introduction of computerized air traffic management applications and new digital communication links between aircraft and ground systems.

Air traffic management communication is therefore transitioning from analog voice communication to digital data communication. Digital networks are already used for air traffic management to a limited extent. The most notable one is the digital Aircraft Communications Addressing and Reporting System (ACARS) introduced in 1978, which uses very high-frequency (VHF), or L-band² satellite channels. In 2001, the European organization for the safety of air navigation (EUROCONTROL) launched the LINK2000+ program to migrate air traffic management applications then running on top of ACARS to the open systems interconnection (OSI) protocol stack on top of the VHF

Digital Link Mode 2 datalink³. However, this OSI protocol implementation is now considered obsolete, and VHF Digital Link Mode 2 suffers from operational problems [4]. The Single European Sky Air Traffic Management Research (SESAR) program will therefore migrate the air traffic management communication infrastructure to a more sustainable technological basis. Broad-band digital datalink technologies for aeronautical communication are therefore under development and will be integrated in an Internet protocol (IP)-based aeronautical telecommunications internetwork, called the future communications infrastructure (FCI). The datalink technologies under development are the Iris satellite link [5], the short-range terrestrial AeroMACS datalink for airport communications [6], and the terrestrial long-range L-band⁴ digital aeronautical communications system (LDACS) [7] that is the focus of this article.

Security and safety are strongly interrelated in aviation, according to Ben Mahmoud et al. [8]: Cybersecurity is therefore an important aspect of the air-ground communication infrastructure. The first aeronautical network introduced, ACARS, originally had no security functions. Then, the Airlines Electronic Engineering Committee developed a standard called ACARS Message Security to provide end-to-end application layer security on top of the existing infrastructure [9]. ACARS Message Security is based on the security specification of the OSI aeronautical telecommunications network, which employs context management middleware for authenticated key establishment [10]. The established keys may then be used by other air traffic management applications without further key exchanges [11]. Context management has been ported not only to ACARS but also to the IP-based aeronautical telecommunications network under development (the FCI) [12]. In addition to application-layer security, the FCI specifies the use of Mobile Internet Protocol version 6 (IPv6) with IP Security (IPSec) [13] and Internet Key Exchange version 2 [14] in the network layer. In the datalink layer, the current datalinks used by ACARS and the OSI aeronautical telecommunications network (VHF and HF channels,

¹ The ground-ground communication infrastructure is outside the scope of this paper.

² The satellite L-band goes from 1,545 to 1,656.5 MHz.

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³ The term "datalink" is used in the civil aviation community to refer to digital communication technologies for air-ground communication related to the safety and regularity of the flight [8].

⁴ The aeronautical L-band goes from 960 to 1,164 MHz.



L-band satellite channels, and VHF Digital Link Mode 2) have no dedicated cybersecurity features [8].

The LDACS datalink will be the primary long-range terrestrial datalink of the future IP-based aeronautical telecommunications network; however, the LDACS specification [15] does not yet cover cybersecurity.

The objective of this article is to discuss possible cybersecurity options for the LDACS datalink, adding layers of security to the IP-based aeronautical telecommunications network. The article focuses on the physical layer, the datalink layer, and the subnetwork layer, because these are the layers implemented in LDACS. Network-layer security is addressed by the IP-based aeronautical telecommunications network protocols, and application-layer security is taken into account by context management. Our discussion of LDACS cybersecurity is based on the major outcomes of a threat analysis. We discuss possible options for implementing the identified security functionalities in the LDACS system and indicate a way forward in the specification of LDACS. Due to the particular properties of the LDACS system (small bandwidth, narrow frequency band, and small packet sizes), it is not feasible to adopt existing security implementation solutions. For example, security approaches of worldwide interoperability for microwave access (WiMAX) standards (e.g., Institute of Electrical and Electronics Engineers [IEEE] Standard 802.16 [16] or IEEE Standard 802.11 [17]), which provide the basis for AeroMACS security and security approaches used in the field of vehicular ad hoc networks [18], are not directly portable to LDACS. The security implementation options discussed in this article are therefore novel and specific to the LDACS air traffic control datalink. However, we believe that once the proper place for the implementation of security in the LDACS protocol stack has been identified, well-established algorithms and schemes (e.g., Diffie-Hellman key exchange [19]) should be utilized [20].

DEFINITIONS

In this article, we define the term “security data” as the metadata required to achieve security goals (e.g., integrity and confidentiality) for data transmitted via the LDACS datalink. In the technical implementation, this may cover message authentication codes,

digital signatures, cryptographic keys, initialization vectors, and other cryptographic elements that have to be exchanged between securely communicating parties.

The notions of confidentiality, integrity, and availability are defined according to [21], [22], and the notion of authenticity is defined according to [23]. Confidentiality is defined as the property that information is not made available or disclosed to unauthorized individuals, entities, or processes. Integrity is defined as the property that data have not been altered or destroyed. Availability is defined as the assurance that information and communication services will be ready for use when expected. Authenticity is defined as a property ensuring that the identity of a subject or resource is the one claimed. “Message alteration” refers to unauthorized (accidental or intentional) modification of an authentic message [24]. “Robustness” refers to the ability of a system to function correctly in stressful environmental conditions or in the presence of invalid inputs [25].

The audit process gathers data about activity in the system and analyzes it to discover security violations or diagnose their cause. Analysis may occur offline after the fact or online in real time. In the latter case, the process is usually called intrusion detection [26].

LDACS SYSTEM ARCHITECTURE

In 2003, at the 11th International Civil Aviation Organization (ICAO) Air Navigation Conference, it was agreed that the aeronautical air-ground communication infrastructure has to evolve to provide the capacity and quality of service required to support future air traffic management concepts. The U.S. Federal Aviation Administration and EUROCONTROL, representing the regions experiencing the most pressure to reform their existing air-ground communication infrastructure, initiated the action plan 17 activity to jointly identify and assess candidate technologies for future digital aeronautical communication systems. Action plan 17 came to an end in late 2007, and the final report concluded that none of the investigated technology proposals could be fully recommended in isolation from the other candidates for all airspace domains [27]. However, action plan 17 activities identified several desirable features that the future system should fulfill. Based on these features,

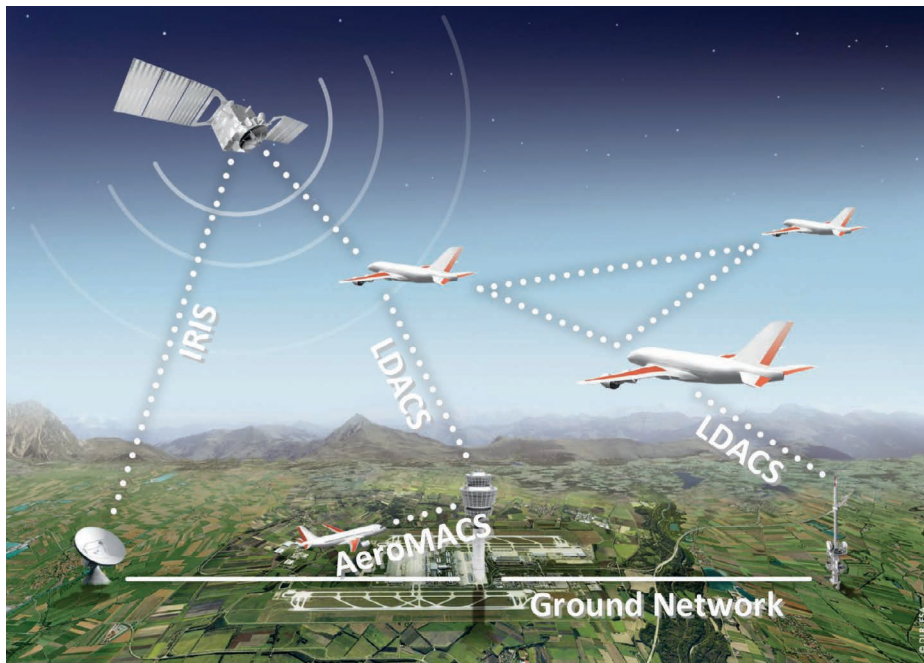


Figure 1.

LDACS subnetwork within the IP-based aeronautical telecommunications network called the FCI. The LDACS subnetwork will be one of several wireless access networks connecting aircraft with the air traffic management networks on the ground. The other datalinks proposed for the aeronautical telecommunications network are the Iris satellite system and the AeroMACS datalink for airport operations.

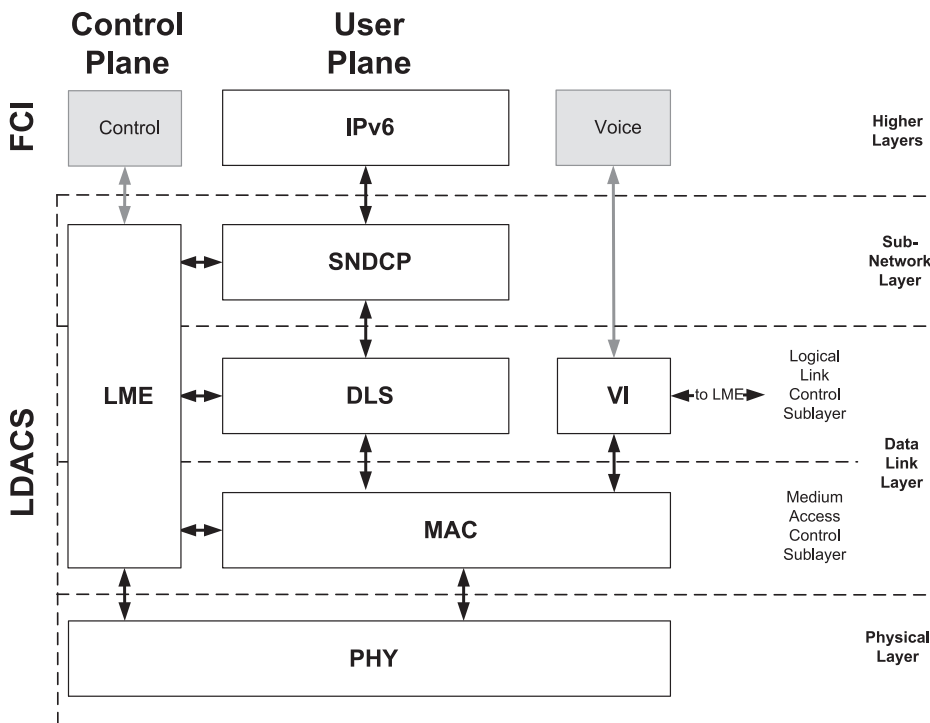


Figure 2.

LDACS protocol stack and functional entities. The interface of LDACS to the IP-based aeronautical telecommunications network called the FCI is the SNDCP. The functional entities within the LDACS datalink are the LME, DLS, voice interface (VI), medium access (MAC) entity, and physical (PHY) layer. The LDACS protocol stack is identical for aircraft and ground station, although the ground station LME and DLS have to be more sophisticated to serve multiple aircraft.

initiatives for the development of an LDACS were started at the end of action plan 17.

One outcome of these initiatives was the technology proposal for a broadband LDACS system [15]. LDACS has been developed by merging aspects of the technologies investigated in action plan 17: broadband aeronautical multicarrier communications [28], wideband data standards P34 (Telecommunications Industry Association standard family TIA-902), and WiMAX IEEE 802.16e technologies [29]. The LDACS datalink is a long-range multi-application cellular broadband system capable of simultaneously providing various kinds of air traffic services and aeronautical operational control communication services via a network of ground stations.

LDACS will be one of several wireless access networks connecting aircraft to the IP-based FCI displayed in Figure 1 [2]. The LDACS subnetwork contains several ground stations, each of them providing one LDACS radio cell. Each ground station is the central entity controlling all air-ground communications within its radio cell [7]. The deployment of the LDACS communication infrastructure is independent of the airspace organization, because LDACS supports transparent handovers between cells.

The LDACS protocol stack defines three layers, the physical layer, the datalink layer, and the subnetwork layer, as illustrated in Figure 2 [30]. The physical layer (PHY in Figure 2) provides the means to transfer data over the radio channel. The datalink layer is organized in two sublayers: the medium access control sublayer and the logical link control sublayer. The logical link control sublayer manages the radio link and offers a bearer service, with different classes of service to the higher layers. It comprises the datalink services (DLSs) and the voice interface. The medium access control sublayer contains only the medium access (MAC in Figure 2) entity. Cross-layer management is provided by the link management entity (LME). The LDACS system interfaces with the IP-based aeronautical telecommunica-

tions network via the subnetwork-dependent convergence protocol (SNDP) that serves also as LDACS subnetwork protocol.

The LDACS ground station supports bidirectional links to multiple aircraft in its cell. The forward link (ground to air) and the reverse link (air to ground) are separated by frequency division duplex. Forward link and reverse link use a 500-kHz channel each. The bandwidth of these channels has been chosen to deploy LDACS in the gaps between the channels of the existing aeronautical distance measuring equipment (DME) system that is used for navigation [31]. The ground station transmits a continuous stream of orthogonal frequency-division multiplexing (OFDM) symbols on the forward link. In the reverse link, aircraft are separated in time and frequency using a combination of orthogonal frequency-division multiple access and time-division multiple access. Aircraft thus transmit discontinuously on the reverse link, with radio bursts sent in precisely defined transmission slots allocated by the ground station.

The LME takes care of the configuration, resource management, and mobility management of aircraft connected to a LDACS ground station. It implements the control plane of the LDACS protocol. Mobility management provides support for registration and deregistration (cell entry and cell exit) and handover between ground stations. Cell entry is requested by an aircraft LME by transmitting a cell entry request message to the ground station LME. If cell entry is possible, the ground station replies with a cell entry response message containing the correct configuration for the aircraft. If several ground stations are available, the LME of the ground station takes care of handovers. LDACS employs a ground station-based handover strategy. That is, handovers are triggered by the current ground station of the aircraft via a management message conveying the correct configuration for the new ground station. Resource management provides closed loop link maintenance and resource allocation through the exchange of control messages. It supports the maintenance of transmit power, frequency adjustment, and time advance of the aircraft's physical layer. Forward link and reverse link resources are assigned by the ground station LME for the transmission of user data.

The DLS sees to the reliable transmission of user data between the aircraft and the ground station. It implements the user plane of the LDACS protocol. It breaks higher-layer packets into fragments for transmission to reduce the probability of errors during the transmission. If an error is detected, only the erroneous fragment is retransmitted by the DLS.

The SNDP provides an adaption layer between the LDACS user plane and the network layer. Although LDACS has been designed for IPv6, it needs to be able to support other network protocols as well. Therefore, all functions related to the mapping of network-layer packets to LDACS-specific subnetwork packets are carried out by the SNDP.

LDACS SECURITY REQUIREMENTS

There is an ongoing discussion in the literature [14], [32] concerning which security functions should be implemented in the different layers of the FCI. The ICAO manual for the aeronautical telecommunications network [14] proposes placing most security

functions in the network layer or above, and this is the approach followed by ACARS and the OSI aeronautical telecommunications network. In [32], an evolved IP-based aeronautical telecommunications network security concept with focus on IPSec over IPv6 is proposed. The concept of [32] also foresees security functions in the air-ground datalinks implementing the physical layer and data-link layer of the aeronautical telecommunications network.

In a similar vein, we argue that security functions implemented in the LDACS subsystem can contribute meaningfully to the overall security of the aeronautical telecommunications network by adding layers of security.

The rationale for providing lower-layer security within the aeronautical telecommunications network protocol stack is to provide a layer of security between the aircraft and the ground station that is independent of higher-layer security functions, thus contributing to the defense-in-depth of the future aeronautical telecommunications network [33]. This lower-layer protection is also invariant under changes of higher-layer protocols. A lower-layer approach also offers the possibility to detect compromised datalink security at an early stage, i.e., before higher layers process any data, and hence initiate counteractions quicker. Furthermore, our proposed lower-layer approach is complementary to higher-layer approaches, because it creates the opportunity to protect the control data of the datalink. In contrast to user data, control data are only exchanged between LDACS devices and are never forwarded to the higher layers. Hence, it is impossible for higher-layer security approaches to contribute to the protection of this control data.

Below, we revisit the outcomes of our threat analysis of LDACS as a subsystem of the FCI [34]. The analysis and our revision are aligned with the common criteria process [35] and build on the work of [21], [32], and [36]. Here, we focus on the LDACS security objectives and classes of security functional requirements derived from them. These objectives and functional security requirements form the basis of the cybersecurity implementation options discussed in the next section.

The basic security objectives for LDACS as a subsystem of the FCI are as follows⁵:

- Objective 1: The operation of the LDACS system security functions shall not diminish the ability of the LDACS system to operate safely and effectively.
- Objective 2: The LDACS system shall support reliability and robustness to mitigate denial-of-service attacks (e.g., deliberate insertion of RF interference).
- Objective 3: The LDACS system shall support message authentication and integrity to prevent message alteration attacks.
- Objective 4: The LDACS system should support confidentiality to mitigate eavesdropping.
- Objective 5: The LDACS system shall support entity authentication to mitigate impersonation attacks.

⁵ The word “shall” denotes a mandatory objective, while “should” denotes a guideline or recommendation [43].

Objective 4 is given only with a “should.” This is mainly because the potential safety impact of eavesdropping on air traffic management services has been assessed with lower severity. Furthermore, safety risks may occur when an aeronautical stakeholder is not able to decrypt air traffic management data. Besides this, the current ICAO policy prevents end-to-end encryption of air traffic service communications ([32], pages 8–14) implying that the application of encryption mechanisms may be restricted in some cases [34], [36].

The following classes of functional security requirements are derived to cover the security objectives defined above [34], [35]:

- ▶ **F_Audit:** Security audit, i.e., recognizing, recording, storing, and analyzing information related to security-relevant activities
- ▶ **F_Auth:** Identification and authentication, functioning to establish and verify attributes, e.g., the identity claimed by an entity
- ▶ **F_Crypto:** Cryptographic support using cryptographic functionality to help satisfy several security objectives
- ▶ **F_Data:** Data protection policies and mechanisms for access control and information flow control inside a device
- ▶ **F_Robust:** Robustness support based on physical layer functionality
- ▶ **F_SecFunc:** Protection of the security system functionality (related to the integrity and the management of the mechanisms that constitute the security system functionality)

Table 1 shows which security objectives are addressed by the listed functional security requirements.

LDACS SECURITY IMPLEMENTATION OPTIONS

The first step in adding security functionality to LDACS is to analyze possible options to implement the identified security functions within the existing system architecture. We restrict ourselves to options that can be added to the existing protocol specification. A design process from scratch might yield different results but is outside the scope of this article. In this section, we investigate implementation options at several points within the LDACS protocol stack. In addition, we discuss the implications of these options.

Security is not free and requires the dedication of resources to its purpose. The second step in the analysis is thus to map the most resource efficient option (in terms of resilience and overhead) that fulfills all requirements for the LDACS protocol stack. This step is carried out in the next sections.

SECURITY IN THE PHYSICAL LAYER

According to resolution 417 of the World Radiocommunication Conference in 2007 [37] modified in 2012 [38], the frequency band 960–1,126 MHz is conditionally allocated to LDACS on a secondary basis. The primary user of this frequency band is the DME radio navigation system.

Table 1.

Security Objectives and Functional Security Requirements	
Security Objectives	Functional Security Requirements Addressing the Objective
Objective 1: Safety and efficiency	F_SecFunc, F_Data
Objective 2: Reliability and robustness	F_Robust, F_SecFunc
Objective 3: Message authentication and integrity	F_Crypto, F_Auth
Objective 4: Confidentiality	F_Crypto, F_Data
Objective 5: Entity authentication	F_Crypto, F_Auth

NOTE: These requirements are not necessarily disjointed. Some requirements imply others; e.g., authentication (F_Auth) requires cryptographic support (F_Crypto). In addition, Objective 2 (robustness against denial of service) and the derived functional security requirement F_Robust cannot be provided by cybersecurity means alone. Additional security (noncybersecurity) means will be required in a comprehensive security concept for LDACS.

LDACS will be deployed in an inlay approach between DME channels. Inlay deployment foresees LDACS channels (500-kHz bandwidth) being deployed between neighboring DME channels (1-MHz spacing) [31], as illustrated in Figure 3a. If larger gaps between DME channels are available, they are preferred. The feasibility of the inlay approach has been demonstrated in flight experiments as shown in Figures 3b and 3c [39], [40].

The specification of the LDACS physical layer is publicly available [15]. This may enable malicious attacks targeting specific features of the system [41]. Attacks on the physical layer aim to disrupt communication between authorized entities. In their simplest form, they aim to increase the noise floor at the receiver and to degrade the signal-to-noise ratio. The consequence is a denial-of-service attack increasing bit, symbol, and packet error rates and hence limiting the throughput of the datalink or causing a temporary loss of connection.

Although the setup of aeronautical infrastructure (cordoned-off airports and flying aircraft, with high security standards and up-tilted antennas) makes interfering with the LDACS signal difficult, additional robustness against denial of service is desirable.

LDACS can be made more robust in the physical layer by one or more of the following approaches:

- ▶ **Increased transmission power:** Although power inefficient, including margins in the link budget calculations of LDACS would help to make the signal more robust against interference.

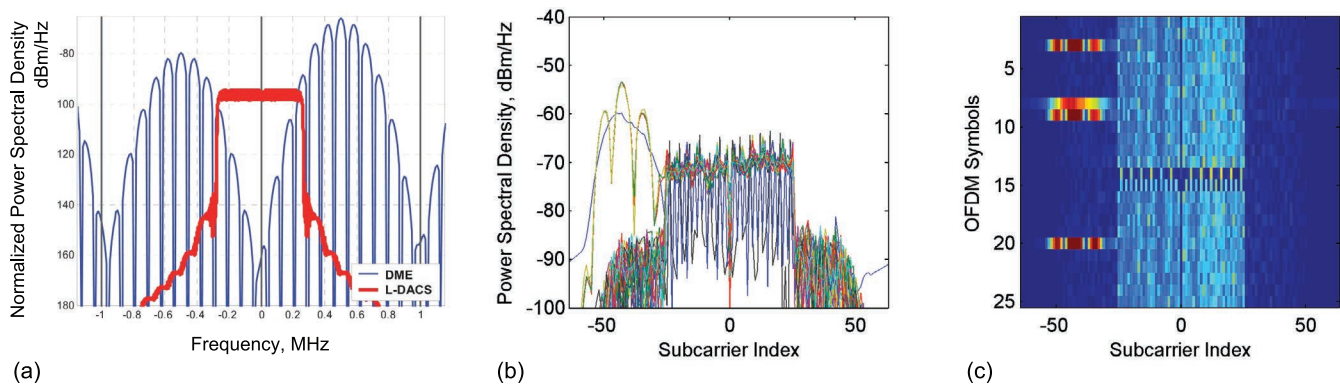


Figure 3.

LDACS inlay concept (left) and recorded LDACS and DME signals from flight experiments [39], [40] (middle, right; courtesy of the authors).

- ▶ **Overlapping cell coverage:** Interfering with the signal of multiple ground stations is more difficult than attacking the signal of one.
- ▶ **Use of location data and directed antennas:** By using surveillance data to acquire location information, the expected angle of arrival can be calculated. Directional antennas can then be focused on the direction from which the signal is estimated to be received. This is particularly useful for aircraft.
- ▶ **Multiple datalinks:** By continuously monitoring a set of features, such as signal-to-noise ratio, packet error rate, location, speed, and Doppler shift, unauthorized signal sources can be detected. If an unauthorized signal is detected, air traffic management communication is switched to a secondary datalink using a different technology, e.g., a satellite datalink.

All of these measures come at significant cost. The robustness of the LDACS signal may be strengthened by increasing the transmission power. However, this measure might cause interference toward the DME system, which is not acceptable. The other measure discussed above requires changes in the infrastructure: multiple ground stations, directed antennas, secondary datalinks, or a combination thereof. The SESAR program is pursuing the multilink concept, i.e., the use of secondary datalinks.

SECURITY IN THE MEDIUM ACCESS CONTROL SUBLAYER

Multiframe Level

LDACS structures the OFDM symbols of the physical layer into multiframe with a 58.32 ms duration, as illustrated in Figure 4. Every four

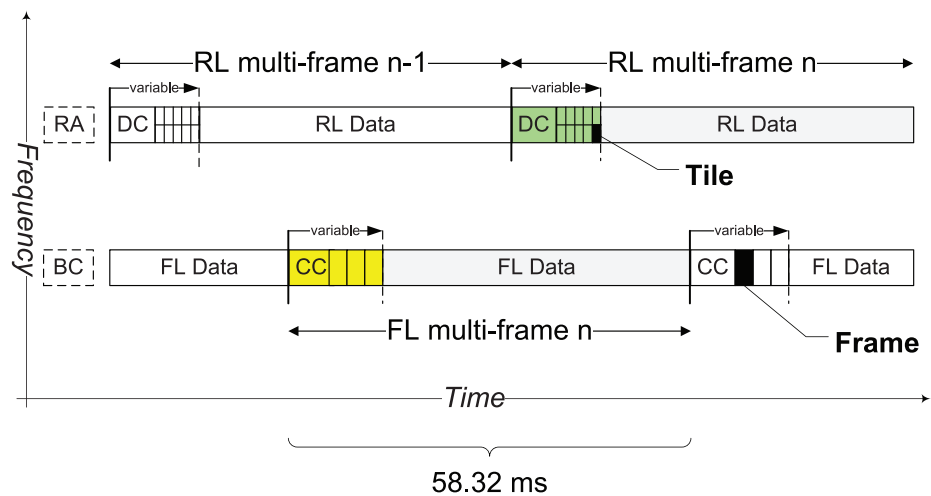


Figure 4.

LDACS frame structure: LDACS uses two radio channels for the forward link (FL; ground to air) and reverse link (RL; air to ground) directions separated by frequency division duplex. Both channels are structured into multiframe of a constant duration of 58.32 ms. Each multiframe is internally structured into a control slot (dedicated control slot (DC) or common control slot (CC) in the figure), and a data slot. The slots are created from frames on the FL and tiles on the RL. The random access (RA) and broadcast control (BC) slots are inserted every four multiframe to create a superframe with a 240 ms duration. FL and RL multiframe are interleaved in time for optimized control slot spacing.

multiframe, an additional random access or broadcast control slot with a 6.72 ms duration is inserted to support the initial cell entry of aircraft. Together this constitutes a superframe with a 240 ms duration.

Introducing security on the multiframe level would require covering the complete multiframe with additional security data. Assuming s -bit security information, the overhead would be $s/19,656$ in the worst case.⁶ Although this approach adds little overhead, it is problematic when it comes to latency requirements, error recovery, and trust relationships.

Multiframe-based security allows security verification on the receiving side only to be performed after the complete multiframe

⁶ Each forward link multiframe comprises 9 OFDM frames. In the most robust coding and modulation scheme, this corresponds to $9 \times 2,184 = 19,656$ bits for the forward link multiframe. Using less robust coding and modulation schemes, the forward link multiframe can grow to 88,992 bits.

has been received. This would prevent the use of both control slots of the same multiframe (spaced 30 ms apart on average; cf. Figure 4) and increase the protocol latency tremendously (leaving out the control slots of the same multiframe, the control slot spacing would effectively increase to 90 ms on average; cf. Figure 5).

Securing multiframe in their entirety also introduces fate sharing between user data packets transmitted in the same multiframe. Error detection and correction (other than the forward error correction done in the physical layer) is performed in the DLS operating on top of the medium access layer. Any bit error in the multiframe will therefore cause the security verification to fail for all packets conveyed in the multiframe. At a worst-case residual bit error rate⁷ of 10^{-5} and multiframe sizes greater than 19,656 bits, this would lead to prohibitive multiframe error rates.

In addition, multiframe-based security limits the applicable trust relationships between a ground station and all participating aircraft. In the forward link, the ground station addresses multiple aircraft in a timed order, one after another. If a whole multiframe was protected as one unit, every involved aircraft had to be able to calculate the integrity verification or the decryption. As a consequence, the utilized cryptographic key material has to be known to multiple aircraft at the same time, which is problematic for symmetric keys. While this is not necessarily a problem for the integrity of user data, it is not possible to transmit confidential user data that is only readable by a single aircraft. In the reverse link, it is not possible to protect complete multiframe, because multiple aircraft contribute data to the multiframe and none of them can protect data on behalf of other aircraft.

Slot Level

Each forward link and reverse link multiframe is divided into two slots: one slot for management and control data (the common control slot on the forward link and dedicated control slot on the reverse link as described in Figure 4) and one slot for the transmission of user data. In addition to these slots, a random access slot and broadcast control slot are inserted every 240 ms for the purpose of cell entries of new aircraft and related management information.

Introducing security at the slot level would require covering complete slots with security data. Because there are two slots per multiframe, this would increase the security overhead vs. the multiframe-based approach. At a minimum slot size of 1 OFDM frame on the forward link, the worst-case overhead ratio would be

⁷ The specified working point for LDACS is a residual bit error rate of 10^{-6} or better after forward error correction. However, LDACS is designed to work up to a residual bit error rate of 10^{-5} .

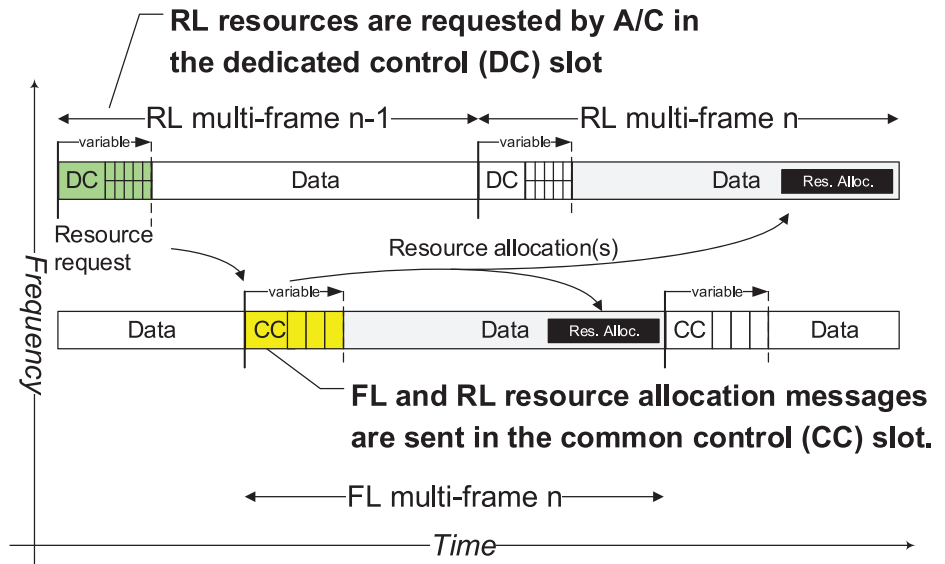


Figure 5.

Data packets are transmitted bundled in a resource allocation (black rectangle). The resource allocation is announced in a table transmitted in the common control (CC) slot. On the forward link (FL), the resource allocation is scheduled locally within the ground station. On the reverse link, aircraft have to request resource allocations in the dedicated control (DC) slot before they can be scheduled by the ground station.

s/2,184. Slot-based security would allow using both control slots of the same multiframe. However, security verification results are only available after each slot has been received completely. With an average medium access delay of 30 ms on the forward link and data slots ending every 58.32 ms, this delay is problematic for digital voice transmissions requiring low latency.⁸

Slot-based security suffers also from the fate-sharing problem. Any bit error in the slot would cause the security verification of all user data packets transmitted in this slot to fail.

Finally, slot-based security would limit the trust relationship between a ground station and all aircraft in a way similar to that of multiframe-based security. Securing the forward link slots might be possible, but protecting the reverse link slots as a unit is not possible for the same reasons as above.

Resource Allocation Level

Resource allocation messages scheduling data packets for transmission on the forward link or the reverse link are transmitted in the common control slot at the beginning of each multiframe according to a transmission request of an aircraft or the transmission backlog of the ground station. Data packets are then bundled into this resource allocation for transmission. This is illustrated in Figures 5 and 6.

Introducing security at the resource allocation level would require covering the transmissions of each user with security data. Since there may be more than one resource allocation per data slot, this would introduce at least the same amount of overhead as slot-based security.

A benefit of security at the resource allocation level is that it has no significant impact on the latency of the LDACS protocol.

⁸ The target latency for voice transmissions is typically <100 ms for the complete processing chain.

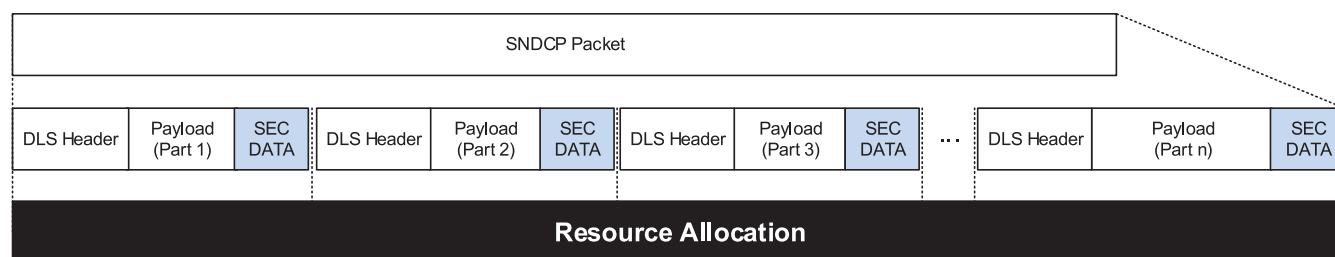


Figure 6.

Security in the DLS. DLS packets are extended with security data (SEC DATA). DLS fragments are transmitted within resource allocations. If a fragment contains errors, it is automatically retransmitted by the DLS in the next resource allocation.

In addition, since resource allocations are assigned per user, individual trust relationships between the ground station and each aircraft become possible.

However, security at the resource allocation level would still introduce fate sharing of packets transmitted in the same resource allocation.

SECURITY IN THE DATA LINK SERVICE

The DLS provides transparent error recovery for the transmission of subnetwork (SNDP) packets. To this end, it breaks packets into fragments of a smaller size and therefore lower error probability, as illustrated in Figure 6. If a bit error is detected in one of the fragments, the DLS will automatically retransmit the erroneous fragment.

Introducing security in the DLS would require covering each fragment with security data. DLS fragments are transmitted in resource allocation and therefore cannot be greater than resource allocations. This approach would thus introduce at least the same amount of overhead as resource allocation-based security. Since DLS fragments are the smallest protocol units processed by the LDACS datalink layer protocol, this is the maximum security overhead.

Security in the DLS has no significant impact on the latency of the LDACS protocol. It allows individual trust relationships to be established between the ground station and each aircraft, and—because security is applied at the fragment level—different security configurations for each application become possible. In addition, it mitigates fate sharing, because security validation and error recovery are both applied at the fragment level.

SECURITY IN THE SUBNETWORK DEPENDENT CONVERGENCE PROTOCOL

The SNDP provides an adaption layer between the LDACS and the network layer. It encapsulates network-layer packets in LDACS-specific subnetwork packets.

Introducing security in the SNDP would require each subnetwork packet to be covered with security data. The amount of overhead introduced by this approach depends on the size of the subnetwork packets, i.e., the size of the network-layer packets to be transmitted.

Security in the SNDP has no significant impact on the latency of the LDACS protocol and allows individual trust relationships to

be established between the ground station and each aircraft and for each application. It is also immune to fate sharing, because error recovery is performed in the layer below by the DLS.

DISCUSSION

The analysis of the possible security implementation options in the existing LDACS architecture leaves us five options to consider. We discuss these options in terms of error resilience, overhead, and security requirements.

QUANTIFYING THE IMPACT OF FATE SHARING

LDACS recovers from residual bit errors after forward error correction by retransmitting erroneous packets. The retransmission mechanism is implemented in the DLS. This implies that user data can contain bit errors at multiframe level, slot level, and resource allocation level, i.e., in the protocol stack below the DLS. A bit error in the user data or the security data at these levels would cause the security verification to fail for all packets covered by the security data. The DLS cannot correct such errors without circumventing security. Therefore, all packets covered by the same security data share their fate with the erroneous packet: security validation failure.

Figure 7 displays the impact of this fate sharing. The lowest failure probability is achieved by adding security data to DLS fragments in which security validation and error detection coincide. The size of the DLS fragments is dynamically adjusted (Label A in Figure 7) to keep the DLS fragment error rate below 5% [30]. At a bit error rate of 10^{-5} , this limits the size of DLS fragments to 641 bytes [42], making fragmentation necessary for larger subnetwork packets. The worst-case failure probability is achieved when covering the complete multiframe with security data. Figure 7 displays the failure probability for multiframe with the strongest coding and modulation (default) and the weakest coding and highest modulation for increased throughput (labeled aggressive coding and modulation in Figure 7).

The results indicate that the increase in failure probability due to residual bit errors (shaded area in Figure 7) is significant. Even with the strongest coding and modulation, the increase in failure probability becomes prohibitive at a residual bit error rate of 5×10^{-6} , well below the design goal of 10^{-5} . For more aggressive coding and modulation, applying security below the DLS is not an option (lightly shaded area in Figure 7).

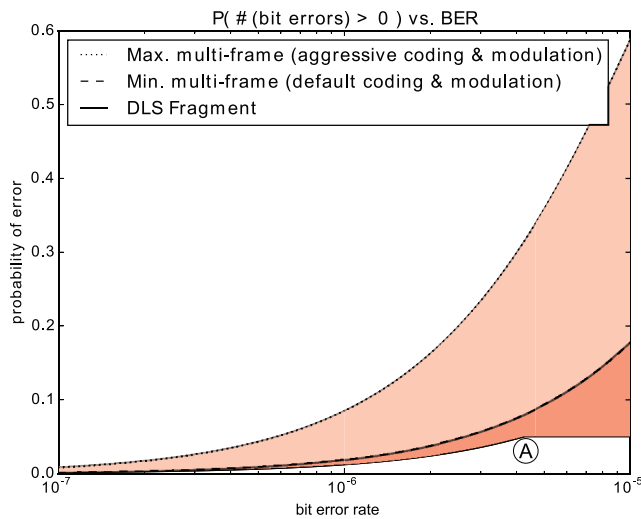


Figure 7.

Impact of fate sharing. The graph displays the probability of the security verification to fail due to residual bit errors. The security verification fails if there is one or more bit errors in the data covered by the security data. The smallest proposed security coverage spans DLS fragments; the largest security coverage spans the entire multiframe. Slot-based and resource allocation-based security lie between these extremes. Error recovery is performed by the DLS by retransmitting DLS fragments. The shaded area identifies the worst-case increase in failure probability due to fate sharing.

QUANTIFYING SECURITY OVERHEAD

Security in the DLS does not suffer from bit errors, because those errors are corrected prior to security validation. However, DLS overhead increases with the bit error rate, because finer fragmentation is required to achieve the desired fragment error rate of 5% or lower. The subnetwork-based approach adds security overhead only once to each packet. It is therefore instructive to investigate security overhead in the datalink level-based approach against overhead in the subnetwork-based approach.

Figure 8 displays the increase of security overhead under an increasing residual bit error rate, assuming $s = 128$ bits of security data per fragment or packet (shaded area in Figure 8). For bit error rates worse than 5×10^{-6} , the security overhead of the datalink layer-based approach increases significantly due to the smaller fragment sizes and more frequent retransmissions.

QUALITATIVE ANALYSIS

Making LDACS more robust in the physical layer requires additional infrastructure: multiple ground stations, directed antennas, or secondary datalinks. Having secondary datalinks as fallbacks is the approach foreseen in the aeronautical telecommunications network by SESAR. However, additional security functionality in the datalink layer is still desirable to build additional layers of defense in depth.

Security at the multiframe level is problematic when it comes to error recovery, latency requirements, and trust relationships in the forward link. The theoretically good overhead ratio does not compensate for these drawbacks. In addition, the reverse link cannot be protected by this approach, because multiple aircraft are

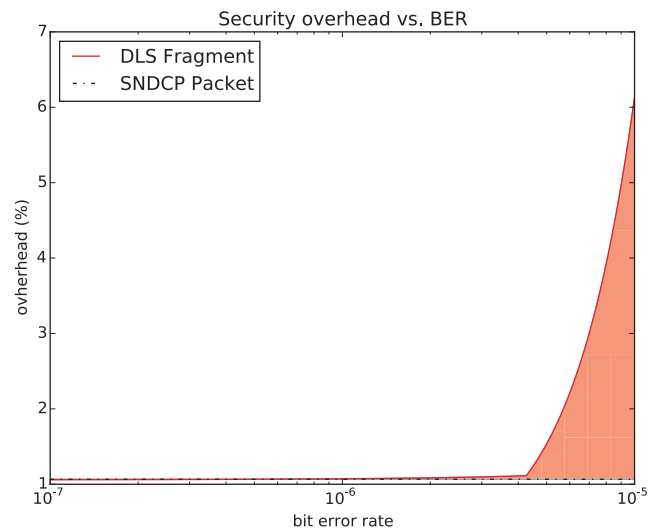


Figure 8.

Security overhead of subnetwork (SNDP)-based security vs. DLS-based security. The shaded area identifies the increase of security overhead due to DLS fragmentation and retransmissions. Nonsecurity overhead is not displayed. For details on nonsecurity overhead, see [42].

consecutively contributing data to the multiframe structure, making it impossible for a single aircraft to secure all data prior to its transmission. The same drawbacks hold for the slot-level approach.

Resource allocations are assigned to each aircraft separately. Security at the resource allocation level would therefore circumvent the limited trust relationship problem. However, like the multiframe-based approach and the slot-based approach, this is not attractive due to large security validation failure probabilities caused by residual bit errors.

Of the two remaining options, security in the DLS and security in the subnetwork layer, the latter is more attractive due to its lower overhead. A minor drawback of this approach is the comparably high placement in the LDACS protocol stack. Security validation can only happen when the subnetwork packet has been completely received. In consequence, security violations are detected later compared to implementation options lower in the protocol stack. However, this does not decrease the level of security. Both approaches support per-application trust relationships as authentication, confidentiality, and integrity are applied on the packet level.

A summary of our findings for all implementation options is displayed in Table 2. Each implementation option is matched against the security objectives derived from the security requirements identified above.

PROPOSED INTEGRATION OF SECURITY FUNCTIONS INTO THE PROTOCOL STACK

The LME sees to the configuration, resource management, and mobility management of aircraft connected to a LDACS ground station. Therefore, we claim that it is the appropriate place in the protocol stack to place the security management functions ($F_{Sec-Func}$). The LME has access to all sublayers of the system; therefore, it is natural to add the auditing function (F_{Audit}) here. The

Table 2.

Match of User-Plane LDACS Cybersecurity Implementation Options and Security Objectives					
	Multiframe	Slot	Resource Allocation	DLS Fragment	SNDCP Packet
Objective 1: Safety and efficiency	Security overhead per multiframe	Security overhead per slot	Security overhead per resource allocation	(-) Maximum overhead: Security overhead per fragment	Security overhead per packet
	(-) Increased latency	(-) Increased latency	No influence on latency	No influence on latency	No influence on latency
Objective 2: Reliability and robustness	(-) Fate sharing per multiframe	(-) Fate sharing per slot	(-) Fate sharing per resource allocation	No fate sharing	No fate sharing
Objective 3: Message authentication and integrity	(-) Forward link only	(-) Forward link only	Forward link and reverse link	(+) Forward link, reverse link, and application	(+) Forward link, reverse link, and application
Objective 4: Confidentiality	(-) Forward link only	(-) Forward link only	Forward link and reverse link	(+) Forward link, reverse link, and application	(+) Forward link, reverse link, and application
Objective 5: Entity authentication	(-) Forward link only	(-) Forward link only	Forward link and reverse link	(+) Forward link, reverse link, and application	(+) Forward link, reverse link, and application
Summary	(-)(-)(-)(-)(-)	(-)(-)(-)(-)(-)	(-)	(-)(+)(+)(+)	(+)(+)(+)

NOTE: Critical limitations are marked with (-); significant advantages are marked with (+).

LME's control over the cell entry procedure indicates that it is the correct place to perform authentication (F_Auth).

The SNDCP offers virtually error-free point-to-point communication between aircraft and ground stations. It does not suffer from the fate-sharing problem, and the added security overhead is low. Therefore, we claim that it is the appropriate place in the protocol stack to apply data protection (F_Data) and cryptography (F_Crypto).

Our proposed assignment of security functions is displayed in Figure 9. We also recommend supporting datalink layer cybersecurity with measures to mitigate denial of service on the physical layer through the SESAR multilink concept (F_Robust).

FUTURE WORK

The main limitation of the analysis presented in the article is that it focuses on the LDACS user plane and does not propose detailed solutions for the LDACS

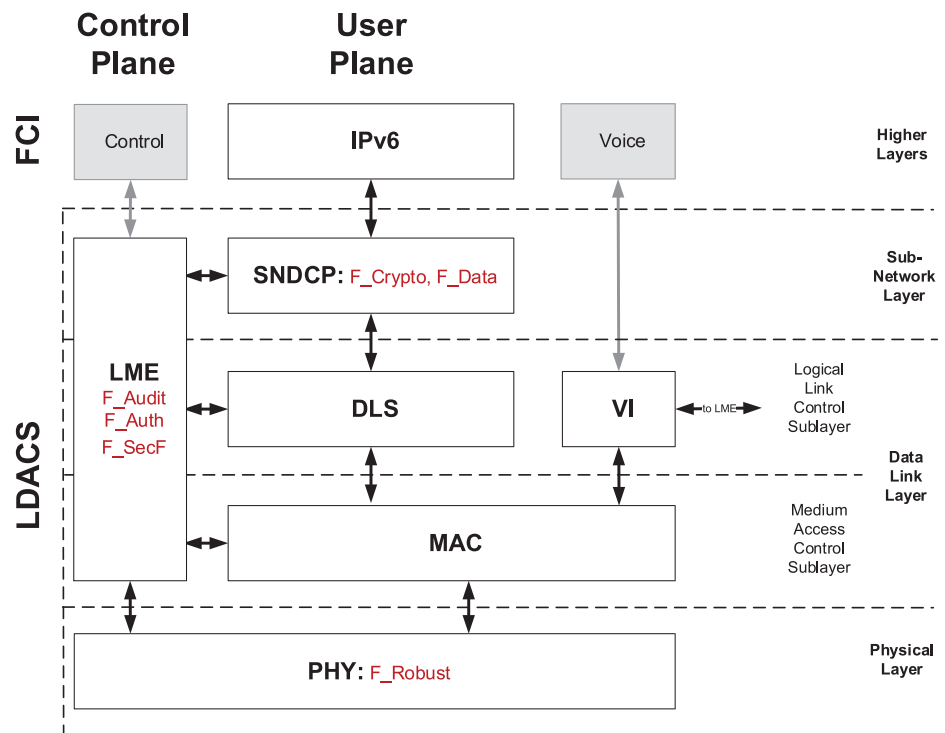


Figure 9.

Proposed mapping of security functions to the LDACS protocol stack.

control plane. Not protecting the control plane leaves a potential security risk open. However, while the integrity protection of control messages would contribute to the security of the system, it is not obvious whether the absence of this protection introduces new threats to the system. The dedicated control and common control slots in each multiframe are primarily used for resource requests and resource allocations to aircraft, which contribute to the medium access-related part of the system's control data. We assume that tampering with resource management information could lead to availability problems caused by altered allocations. However, the likelihood and the impact of such a complex attack, which would have to be based on faked management information, have not been analyzed so far.

CONCLUSION

To protect air traffic management information exchanged via the aeronautical telecommunications network, multiple levels of security will have to be implemented. LDACS can contribute security functions in the datalink layer and the physical layer to the defense-in-depth of the future aeronautical telecommunications network.

A key finding of this work is that the user plane of the current LDACS protocol design provides sufficient resources to add security data with acceptable overhead. This is also reflected by our recommended cybersecurity implementation in the LME and the subnetwork protocol. We believe that not securing the control channel does not introduce new threats; however, authentication and integrity protection of the control data would aid in the detection of attacks and hence potentially improve the system's availability. Future research will be necessary to investigate the threats in case of an unprotected control channel and to clarify whether LDACS offers options to secure the control channel with combined control plane and user plane security data.

The way forward leads through the LDACS standardization effort that has been started by the ICAO communication panel in late 2016 and is supported by SESAR Project 14, "Essential and Efficient Communication Navigation and Surveillance Integrated System." The findings of this article will be provided as input to these activities. ♦

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